

# Retrieve, Locate, Generate: An Oracle-Substitution Diagnostic for Literature-Grounded QA

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## Abstract

Literature-grounded question answering requires a system to retrieve the right paper, locate the right evidence within it, and generate an answer faithful to that evidence. These are three separate stages, each with its own failure mode. We present a stage-wise diagnostic study of this pipeline on LITTRACEQA, a shared task requiring paper retrieval, coarse evidence localization (table/figure/page), and structured answer generation. Our pipeline combines cross-model hypothesis diversity for retrieval, a filter-then-rank evidence localizer (caption-pattern match, embedding similarity, evidence-type gate), and cross-model semantic entropy for generation-time answer aggregation and confidence estimation. We use an oracle-substitution ablation ladder, successively handing the pipeline the gold paper and then the gold evidence value, and reading off the resulting change in each stage’s own metric. These are sequential oracle-intervention gains, not an additive error decomposition: the stages are measured with different quantities and are not independent. Read that way, the ladder points at evidence localization as the dominant remediable bottleneck — multiple-choice accuracy rises from 0.42 with predicted evidence to 0.76 with the gold evidence value, and end-to-end accuracy stays near 0.29 even on the questions where retrieval already returned every gold paper. We also test whether cheap uncertainty signals computed at retrieval time predict which stage will fail before it does. Along the way we surface two infrastructure-level findings with direct implications for anyone building literature-grounded QA systems. First, evidence page numbers in this and similar benchmarks are keyed to a paper’s canonical published version rather than its arXiv preprint: a real failure mode when a paper has been substantively revised, though a systematic paired ablation across 45 questions finds it is an occasional rather than pervasive hazard in this pool. Sec-

ond, a substantial fraction of papers hosted on OpenReview (ICLR/ICML/NeurIPS) are unreachable by ordinary scripted HTTP requests due to a JS bot-challenge, a venue-correlated failure mode unrelated to any modeling choice. We report our diagnostic protocol, quantitative results with bootstrap confidence intervals given the benchmark’s small public dev set, and a qualitative failure taxonomy, framing this as a resource and lessons-learned contribution rather than a leaderboard claim. All multiple-choice numbers are a best-case diagnostic condition: they assume access to the gold option text, which a genuine blind submission would not have.

## 1 Introduction

Grounding a language model’s answer in retrieved documents is usually evaluated with a single end-to-end accuracy number. But a literature-grounded QA pipeline is not one system; it is at least three: a retriever that must find the right paper, a localizer that must find the right evidence within that paper, and a generator that must produce an answer faithful to that evidence. A low end-to-end score is consistent with several different underlying stories: the retriever missed the paper, the localizer missed the table, or the generator misread a correctly-located table, and each of these calls for a different fix. Treating the pipeline as a black box conflates them.

This paper reports a stage-wise diagnostic study of literature-grounded QA on LITTRACEQA, a shared task at the GroundLM 2026 workshop that requires paper retrieval, coarse evidence localization (identifying a supporting table, figure, text span, citation, or equation, down to a page and, where applicable, a visible label), and answer generation in one of three formats (multiple-choice, freeform, or a structured table). We build a pipeline combining cross-model hypothesis diversity for retrieval (extending HyDE-style

hypothetical-document embedding (Gao et al., 2023a) across several heterogeneous LLMs), a filter-then-rank evidence localizer, and a cross-model semantic-entropy aggregation step for generation (Kuhn et al., 2023), then use an *oracle-substitution ablation ladder* — sequential oracle interventions read in each stage’s own metric, not an additive error decomposition — to attribute end-to-end error across stages rather than reporting only the end-to-end number.

Two further findings, surfaced while simply getting the pipeline running, turn out to be central rather than incidental. First, evidence locators in this benchmark are keyed to a paper’s *canonically published* version rather than its arXiv preprint; the two can differ enough in content and pagination that naive arXiv substitution silently breaks evidence grounding. Second, papers hosted on OpenReview (ICLR, ICML, NeurIPS) are unreachable by ordinary scripted HTTP requests, including the official API, due to a JavaScript bot-challenge, while ACL Anthology, CVP, and ECCV sources are not. Both mean a system’s measured accuracy on this task family can be shaped by *which venue a question’s paper comes from*, before any modeling choice is made — exactly the confound a stage-wise diagnostic should surface.

Our contributions: (1) a literature-grounded QA pipeline extending cross-model retrieval-diversity and cross-model uncertainty methods to a new grounding task; (2) an oracle-substitution diagnostic protocol that attributes end-to-end error across retrieval, localization, and generation through sequential oracle interventions (not an additive decomposition), with bootstrap CIs given the benchmark’s 55-example dev set; (3) a test of whether retrieval-time uncertainty predicts which stage will fail; and (4) two infrastructure findings — canonical-source version drift and venue-correlated fetch failure — that likely generalize beyond this benchmark.

## 2 Task: LitTraceQA

LITTRACEQA asks a system to answer a research question grounded in a pool of 27,487 candidate papers (ACL, EMNLP, NAACL, NeurIPS, ICLR, ICML, CVPR, ICCV, ECCV; 2024–2025), described only by title, authors, abstract, venue, and year. For each question the system must return (a) the supporting paper or papers; (b) coarse evidence locations, meaning a paper id, an evidence

type (table, figure, text span, citation context, or equation/algorithm), a page, and for tables/figures a visible label such as “Table 4”; and (c) an answer in one or more of three formats: multiple choice, freeform text, or a structured table matching a per-question schema. The public development split has 55 questions (26 single-paper, 29 multi-paper; primary evidence type: 17 table, 15 figure, 9 text span, 7 citation, 7 equation), each with a gold paper set, gold evidence, and gold answer(s). The official local evaluator reports macro-averaged precision/recall/F1 for paper retrieval (exact paper-id set match) and for evidence (exact match on a normalized 4-tuple of paper id, evidence type, page, and visible label), plus multiple-choice accuracy, freeform exact match (case/whitespace-normalized, no numeric tolerance), and table row-F1 and cell accuracy. These are five independent metric families with *no combined score*, which we take as an implicit endorsement of exactly the stage-wise view this paper takes.

One schema gap is worth flagging up front. The input-only question record available to a real (hidden-test) participant includes the question and its required answer format(s), but *not* the multiple-choice option text, which appears only in the gold file. For our dev-set diagnostic experiments we surface the gold options as generation-time context, since there is no other way to produce a gradable letter, but this means our multiple-choice numbers are a best-case estimate under an assumption (knowledge of the options) that would not hold for a genuine blind submission on hidden test data. We return to this in §8.

### 2.1 Canonical-source grounding: a real but apparently rare failure mode

Early on we traced one gold example to its source. Paper ac12025\_00005’s gold evidence cites “Table 4” on p.6 for a NaturalQuestions comparison; the arXiv v1 preprint contains no such table or comparison at all (it is an earlier, differently-scoped version), while the ACL Anthology PDF linked from the task’s own metadata pool reproduces the gold locator exactly.

We then ran a systematic paired ablation: for the 52 of 70 unique gold papers we could match to an arXiv entry by normalized title (the pool’s arxiv\_id field is null throughout, so matching was manual), we localized on both arXiv and canonical PDFs over the resulting 45-question subset. The anecdote does not scale: 42 of 45 ques-

tions score *identically*, mean difference +0.024 favoring arXiv, bootstrap 95% CI  $[-0.011, 0.076]$  (crosses zero); only 3 change (2 up, 1 down). ac12025\_00005 itself is excluded — OCR-style spacing in its title (“500x C ompressor”) broke matching.

Version drift is thus a *real* but occasional failure mode: it can silently break grounding on a substantively revised paper, but is not systematic across this pool. The practical takeaway is unchanged — use the canonical source, which is free — but the strength of the claim is now calibrated by data rather than one example.

## 2.2 OpenReview’s bot-challenge as a venue-correlated failure mode

Fetching the canonical PDF for all 70 unique gold papers on 2026-07-04 with a scripted Python requests GET, 32 succeeded (ACL Anthology, CVF, ecva.net); the remaining 38 — every one on OpenReview (ICLR, ICML, NeurIPS) — returned HTTP 403 with a “Verifying your browser” challenge, including via api2.openreview.net. A headless-browser session (Playwright/Chromium) recovers all 38: one navigation sets a session cookie, after which same-session fetches succeed under a  $\sim 60$ -req/min limit. This can depend on date, endpoint, IP reputation, and request rate, so we treat it as a venue-correlated confound observed under these conditions rather than a fixed OpenReview property: ICLR/ICML/NeurIPS scores can be depressed for reasons unrelated to retrieval or localization, and reproducers should budget for a non-scripted path.

## 3 Method

### 3.1 Retrieval: cross-model hypothesis diversity

Research questions and paper abstracts are lexically distant even when topically related, which is exactly the vocabulary-mismatch setting HyDE-style retrieval targets (Gao et al., 2023a). We extend this idea across models: for each question,  $N=4$  heterogeneous LLMs spanning distinct model families (qwen3-235b-instruct, llama4-maverick, gemma4-31b, and granite41-30b) each generate a short hypothetical passage that would answer the question. Each hypothesis is embedded (BGE-large-en-v1.5), and the  $L_2$ -normalized centroid of the  $N$  embeddings is used as the query

vector against a FAISS IndexFlatIP index over the paper pool’s title+abstract embeddings. We compare this (CMHA) against three baselines: a single model’s HyDE hypothesis (**single\_hyde**), dense retrieval on the raw question with no hypothesis generation (**dense\_raw**), and lexical retrieval (**BM25**). We additionally record each query’s *diversity score*, the mean pairwise cosine distance between the  $N$  hypothesis embeddings, as a candidate predictor of downstream difficulty (§4).

### 3.2 Localization: filter-then-rank evidence grounding

The evidence type required for a given question (table, figure, text span, citation context, or equation/algorithm) is given as part of the question record, so localization reduces to a simpler problem: given a paper’s canonical full text (§2.1), find the right page and, for tables/figures, the right visible label. The localizer combines three signals: (i) a caption-pattern match (regex for Table N / Figure N mentions, normalized to the evaluator’s label rule); (ii) embedding similarity (BGE-large-en-v1.5) between the question and each candidate’s text; and (iii) an evidence-type gate that restricts table/figure questions to pages carrying a same-type label. For text/citation/equation questions we rank whole pages by (ii) and return the top page; for table/figure questions we rank same-type labels by (ii) over the caption’s surrounding text, falling back to whole-page ranking when the paper has no matching label.

### 3.3 Generation: cross-model semantic-entropy aggregation

Given the located evidence, the same 4-model ensemble independently produces an answer in the format(s) the question requires, conditioned on the evidence text as context. We aggregate the four answers per format. For freeform answers, we cluster the four responses by bidirectional NLI entailment, following the semantic-entropy criterion of Kuhn et al. (2023) but across models rather than across samples from one model. Using cross-encoder/nli-deberta-v3-base (run on CPU), an ordered pair  $(a, b)$  counts as entailment when the softmax entailment probability is  $\geq 0.5$ ;  $a$  and  $b$  are equivalent if their normalized strings match or entailment holds in both directions. Equivalent pairs are merged transitively by union-find, so an intransitive triple still

collapses to one cluster. The final answer is the lowest-index member of the largest cluster (ties broken toward the earlier model in the fixed order `qwen3`  $\rightarrow$  `llama4`  $\rightarrow$  `gemma4`  $\rightarrow$  `granite41`); for multiple-choice we cluster by exact letter match instead. In both cases we report the Shannon entropy of the resulting cluster-size distribution as a cross-model confidence signal: low entropy means the models agree, and high entropy means they disagree, which we treat as a candidate abstention trigger. For table answers we take the longest non-empty per-model row list as a simple majority-style heuristic, which we do not consider a principled solution (§8).

#### 4 Diagnostic Protocol: Oracle-Substitution Ablation Ladder

An end-to-end accuracy number cannot tell us whether a wrong final answer came from retrieving the wrong paper, localizing the wrong evidence within the right paper, or misreading correctly-located evidence. We probe this by substituting oracle (gold) information at each stage and re-measuring that stage’s own metric. These are *sequential oracle-intervention gains*, each read in its own metric; paper-F1, evidence-F1, and answer accuracy are different quantities and the stages are not independent, so we do not treat the rungs as additive components of a single error rate.

- **Retrieval rung:** CMHA-predicted papers, scored on paper precision/recall/F1 and on the retrieval-availability metrics of Table 5 (Recall@ $k$ , MRR, exact gold-set recovery); no evidence or answer attempted at this rung.
- **Localization rung:** *gold* papers handed to the localizer, scored on evidence P/R/F1. Because the paper is gold by construction, any evidence error here is attributable to localization alone, not retrieval.
- **Generation rungs:** gold papers and either (a) the located (possibly wrong) evidence from the localization rung, or (b) the *gold evidence value*, handed to the generator, scored on multiple-choice accuracy, freeform exact match, and table metrics. Comparing (a) against (b) bounds how much generation-stage error is inherited from upstream localization error.

**What the generator actually receives.** In condition (a) the generator is given, per located evidence item, the extracted text of the located PDF page, parsed with PyMuPDF and truncated to a 1,500-character slice that the prompt builder re-caps to 1,200 characters per item (App. A). In condition (b) it is given the benchmark’s `evidence_text_or_value` string verbatim: a short gold snippet (e.g. a single cell value “15.66”, one quoted sentence, or one bibliography line), *not* a reconstructed table with row/column structure, a full page, or a figure image. No figure is ever passed as pixels; equations reach the model only as whatever the PDF text layer contains. Condition (b) is therefore best described as “generation given the gold coarse locator value”: it is an upper bound on what localization could hand downstream under the current evidence representation, not an oracle for answer-sufficient context. Where the gold value alone under-determines the answer (§5, `q_001`), residual error in (b) is shared between evidence representation and generation, and we do not attribute it to generation alone. The two conditions also differ in serialization and token budget (raw page text vs. a short gold string). To check that this difference is not what drives the gold-vs-predicted gap, we add a rung (Table 2, row 4) that gives the generator the gold value *plus* the text of its gold-locator page, in the same page-text window condition (a) uses. This *raises* MC accuracy to 0.829 (from 0.756) and leaves freeform EM statistically unchanged (0.346, CI [0.200, 0.474]); it does not fall toward the predicted-evidence rung (0.415). The serialization / budget difference is therefore not masking a generation deficit — if anything, real page context helps — and the localization  $\rightarrow$  generation drop is attributable to the localizer selecting the wrong page or label, not to how evidence is rendered.

Given the public dev set’s 55 questions, we report all metrics with percentile bootstrap 95% confidence intervals (2000 resamples of queries with replacement) rather than bare point estimates, alongside qualitative failure cases; §8 details the resulting adaptation caveat (these same 55 questions also shaped the pipeline’s design).

## 5 Results

Table 1 is directionally consistent with the retrieval-stage hypothesis: CMHA outperforms

| Retrieval config  | Paper F1 (macro) |
|-------------------|------------------|
| CMHA (ours)       | 0.199            |
| Single-model HyDE | 0.180            |
| BM25              | 0.168            |
| Dense (raw query) | 0.160            |

Table 1: Paper-retrieval sanity check, top- $k=5$ , macro paper-F1. With a fixed five-paper prediction set this metric penalizes precision even when a gold paper is retrieved (it caps at 0.33 for a single-gold question), so it understates retrieval quality; Table 5 reports availability-oriented metrics that track downstream usefulness. Point estimates only; the paired bootstrap below is more cautious.

single-model HyDE, which in turn outperforms both non-HyDE baselines (top- $k=5$ ); these are point estimates, and the paired test below is more cautious.

A paired bootstrap on per-query paper-F1 (shared query resamples, 2000 draws) tests this directly: CMHA’s gain over single-HyDE (+0.018, CI  $[-0.010, 0.053]$ ), same-model-x4 (+0.023, CI  $[-0.015, 0.063]$ ), and BM25 (+0.031, CI  $[-0.013, 0.075]$ ) all cross zero, not significant at  $n=55$ ; only the gain over dense retrieval on the raw query (+0.039, CI  $[0.003, 0.075]$ ) is significant. Four same-model samples ( $T=0.7$ , qwen3-235b-instruct) reproduce neither CMHA’s diversity (0.019 vs. 0.086,  $\sim 4.6\times$  lower) nor a significantly different paper-F1; three independent CMHA reruns give 0.199, 0.203, 0.196 (std 0.003), confirming the point estimate is stable.

**Retrieval availability, not just F1.** Macro paper-F1 on a fixed top-5 set conflates two things a downstream stage cares about separately: whether a gold paper is *present* in the candidate set, and how much precision was spent to get it there. Re-scoring the same predictions with availability-oriented metrics (App. B, Table 5), CMHA recovers a gold paper in its top 5 on 63.6% of questions (Hit@5) and the *entire* gold paper set on 38.2% (SetRecall@5), both ahead of every baseline; MRR 0.492. The split by task family is the more important story: CMHA’s Hit@5 is 0.808 on single-paper questions, but on multi-paper questions Recall@5 — the fraction of required papers actually retrieved — is only 0.150. Multi-paper retrieval, not multi-paper localization, is where most of the multi-paper evidence-F1 is lost (§5.3), and on roughly 62% of dev questions the downstream

stages never even see all the papers they would need.

Table 2 reports the full oracle-substitution ladder with bootstrap 95% CIs. Two patterns stand out. First, evidence localization given the *correct* paper still only reaches evidence-F1 = 0.217, barely above the retrieval rung’s paper-F1 and far from ceiling. Second, generation given the *gold evidence value* reaches 0.756 multiple-choice accuracy but only 0.385 freeform exact match. A concrete case (q\_001) shows why: the gold evidence\_text\_or\_value is the single string “15.66”, but the correct answer (“14.70”) is a *derived* quantity that “15.66” alone cannot determine, and the ensemble returns it verbatim. As §4 notes, this is a failure of the gold-value representation as much as of generation, so we read this rung as bounding the localization→generation loss rather than isolating intrinsic generation error.

Comparing four localization strategies on this rung (App. B) shows evidence-F1 = 0.217 is not an artifact of our particular localizer: whole-page dense ranking scores 0.203, BM25 over pages 0.118, a page-1 floor 0. Even given the right paper and a good embedding model, coarse localization on this benchmark tops out near 0.29 on labelled and 0.11 on unlabelled evidence — the localizer is weak, but the task is also hard.

Holding the paper at gold and varying only whether the generator gets the gold evidence value or the localizer’s own (evidence-F1 = 0.217) predictions, MC accuracy falls 0.756 → 0.415 and freeform EM 0.385 → 0.154; the gold and predicted MC CIs ( $[0.654, 0.864]$  vs.  $[0.292, 0.524]$ ) do not overlap, and a paired McNemar test agrees (MC: 16 questions correct only with the gold value vs. 2 only with predicted,  $p = 0.0013$ ; freeform 7 vs. 1,  $p = 0.070$ ). This transition mixes localization with evidence-representation error (§4); either way, the loss is not intrinsic generation reasoning.

**Is the loss upstream of retrieval?** Rather than multiply paper-F1 by answer accuracy (which would treat paper-F1 as  $P(\text{retrieval succeeded})$  and assume stage independence), we condition directly. On the 21 questions where CMHA’s top-5 contains *every* gold paper, the end-to-end top-5 system still scores only MC 0.294 / EM 0.150 / evidence-F1 0.333; on the 23 where the top-1 paper is gold, MC 0.316–0.368. Even handed the right papers, the system loses most of the accuracy the gold-value rung (0.756 MC) shows is reach-

| Rung                                               | Paper F1             | Evidence F1          | MC accuracy          | Freeform EM          | Table row-F1         | Table cell-acc       |
|----------------------------------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| Retrieval (CMHA)                                   | 0.199 [0.164, 0.232] | —                    | —                    | —                    | —                    | —                    |
| Localization (oracle papers)                       | 1.000                | 0.217 [0.140, 0.290] | —                    | —                    | —                    | —                    |
| Generation (gold evidence <i>value</i> )           | 1.000                | 1.000                | 0.756 [0.654, 0.864] | 0.385 [0.235, 0.529] | 0.558 [0.344, 0.770] | 0.293 [0.067, 0.495] |
| Generation (gold <i>value</i> + matched page ctx.) | 1.000                | 1.000                | 0.829 [0.739, 0.920] | 0.346 [0.200, 0.474] | 0.479 [0.303, 0.658] | 0.279 [0.102, 0.442] |
| Generation (predicted evidence)                    | 1.000                | 0.217 [0.140, 0.290] | 0.415 [0.292, 0.524] | 0.154 [0.056, 0.250] | 0.356 [0.150, 0.560] | 0.091 [0.000, 0.200] |
| End-to-end (top-1, no oracle substitution)         | 0.353 [0.259, 0.444] | 0.127 [0.057, 0.189] | 0.268 [0.160, 0.370] | 0.115 [0.000, 0.200] | 0.068 [0.000, 0.150] | 0.000 [0.000, 0.000] |
| End-to-end (top-5, no oracle substitution)         | 0.199 [0.164, 0.232] | 0.127 [0.057, 0.194] | 0.268 [0.160, 0.370] | 0.115 [0.000, 0.200] | 0.068 [0.000, 0.150] | 0.000 [0.000, 0.000] |

Table 2: Oracle-substitution ladder, bootstrap 95% CIs in brackets (2000 resamples of queries with replacement). Rungs are sequential oracle interventions, each read in its own metric, not additive error components. The “gold evidence *value*” rung feeds the generator the benchmark’s short evidence\_text\_or\_value string, not a reconstructed table/page/figure, so it upper-bounds the localization→generation loss rather than isolating generation error (§5); row 4 adds the gold-locator page text in the matched page-text window and shows the gap is not a serialization artifact. MC accuracy assumes gold option text (best-case; §8). Denominators: MC over 41 questions, freeform EM over 26, table metrics over 11. The first five rows share the top-5 CMHA setup of Table 1; the last two use no oracle substitution at any stage (genuine end-to-end), differing only in top-1 vs. top-5 retrieval scope. “—” marks rungs that never produce a table answer.

able — the loss is at localization and downstream, not retrieval.

### Does the cross-model ensemble earn its cost?

On the gold-value rung the four generators alone reach MC {0.683, 0.756, 0.756, 0.707} / EM {0.308, 0.385, 0.346, 0.192}; the semantic-entropy ensemble (MC 0.756, EM 0.385) *exactly matches* the best single model and does not beat it, and no-NLI majority voting is slightly worse (0.732 / 0.346). On predicted evidence the ensemble edges best-single (MC 0.415 vs. 0.366) but at  $n=55$  this is untested. We therefore present cross-model aggregation as a robustness choice, not a demonstrated gain.

The full top-5 end-to-end rung (localize across all five candidates, keep the single highest-confidence locator) gives evidence-F1/MC/EM numerically identical to top-1, though the two agree on the chosen paper for only 28/55 questions — a wider retrieval net changes *which* evidence is chosen without moving downstream accuracy, so its fivefold cost is not paying off. A naive baseline (dense top-1, first three pages, one model, no localization or ensemble) scores paper-F1=0.364, MC=0.220, freeform=0.000; McNemar favours our pipeline on both axes but not significantly at  $n=55$ , though the naive baseline’s total freeform failure — invisible to paper-F1 — is consistent with localization and ensembling doing real work.

## 5.1 Failure Taxonomy

To understand *what kind* of error dominates beyond the aggregate numbers, we hand-read a stratified sample of 20 incorrect predictions from the predicted-evidence generation rung (all 4 cases

| Failure category                                                | Count |
|-----------------------------------------------------------------|-------|
| Localizer resolves to wrong page/label within the correct paper | 10/20 |
| Right table/figure, wrong row/cell                              | 4/20  |
| Over-conservative abstention on multi-paper synthesis           | 3/20  |
| Reference-index lookup failure                                  | 2/20  |
| Evidence-type category mismatch                                 | 1/20  |

Table 3: Hand-categorized failure taxonomy. This is a *deliberately stratified* sample of 20 incorrect predictions — all 4 cases with perfect coarse localization but a wrong answer, plus 16 sampled across formats where localization failed — *not* a uniform draw. The counts show which failure modes recur and must not be read as prevalence rates over the 38 incorrect predictions in the rung.

where the coarse localization metric was perfect yet the final answer was still wrong, plus 16 cases sampled across multiple-choice, freeform, and table questions where localization itself failed), and assign each to one of five failure categories. Table 3 reports the breakdown.

The largest category (10/20) is the localizer landing on the wrong page or label *within the correct paper*: e.g. q\_014 (best API-Bank backbone in AgentNet; gold Table 1, p. 7) gets Table 3 on p. 9, and the generator answers confidently from it. Embedding similarity to the question lets a table answering a closely related but distinct quantitative question outscore the true target. The second category (4/20, and *all four* cases where coarse localization scored perfect) is right table/figure, wrong row or cell: the evaluator’s key (paper\_id, source\_type, page, table\_id) credits localization as soon as the table is named, but on q\_010 (lowest-NRMSE-std system in a 15-row table) the

generator still picks the wrong row. A system can thus score full evidence-F1 and still be unable to answer, because the metric never requires the answer-bearing cell.

The remaining three categories are smaller. Generation *under-using* given evidence (3/20): on multi-paper questions like q\_025 the ensemble hedges to “insufficient information” despite holding usable text spans from all four papers. Reference-index lookup (2/20, q\_003/q\_005): the semantic localizer has no notion of enumerated reference position; a deterministic reference-list parser (regex over [N] markers, author–year fallback) instead answers 3 of the 5 citation-index questions end-to-end (vs. 0 for the shipped pipeline), the two misses needing in-text citation counting over a two-column PDF — an extraction, not a ranking, problem. Evidence-type mismatch (1/20, q\_023): table-derived gold evidence for a citation-comparison question, classified by us as `citation_context`, mismatches on `source_type` alone.

## 5.2 Does cheap uncertainty predict failure before it happens?

A practical motivation for computing retrieval-time diversity and generation-time entropy is routing: if either signal reliably predicts failure, a system could allocate more compute (or abstain) on hard queries cheaply. We treat this subsection as exploratory — the samples are small (41 MC, 26 freeform), all thresholds are read off these same questions, and entropy over four model outputs takes only a handful of distinct values.

Generation-time entropy points the right way but is not, on this evidence, a reliable detector. It correlates negatively with correctness (point-biserial  $r = -0.22$  for MC, bootstrap 95% CI  $[-0.52, 0.09]$ ;  $r = -0.48$  for freeform, CI  $[-0.86, 0.15]$ ), and its AUROC for flagging an *incorrect* answer is 0.65 (CI  $[0.47, 0.82]$ ) for MC and 0.74 (CI  $[0.28, 1.0]$ ) for freeform — both CIs include chance. Using the raw number of semantic clusters instead of the Shannon entropy gives an almost identical AUROC (0.62 / 0.74), so the entropy computation adds nothing over simply counting how many distinct answers the four models gave. What does show through is a usable selective-prediction operating point: answering only when all four models agree (entropy = 0) covers 37% of MC questions at 0.60 accuracy (vs. 0.41 overall) and 19% of freeform questions at

| Group                   | $n$ | Evid. F1 | MC acc. ( $n_{MC}$ ) |
|-------------------------|-----|----------|----------------------|
| <i>By task family</i>   |     |          |                      |
| Single-paper            | 26  | 0.372    | 0.300 (20)           |
| Multi-paper             | 29  | 0.078    | 0.524 (21)           |
| <i>By evidence type</i> |     |          |                      |
| Table                   | 17  | 0.348    | 0.273 (11)           |
| Figure                  | 15  | 0.233    | 0.357 (14)           |
| Text span               | 9   | 0.167    | 0.571 (7)            |
| Equation/algorithm      | 7   | 0.143    | 0.667 (6)            |
| Citation context        | 7   | 0.000    | 0.333 (3)            |

Table 4: Descriptive (non-bootstrapped) breakdown of the predicted-evidence rung by question type.  $n$  counts all questions per group; MC accuracy is over the  $n_{MC}$  questions in that group with a multiple-choice answer type. Evidence-F1 uses all  $n$ .

0.60 accuracy (vs. 0.15); on a risk–coverage curve, MC selective accuracy is 0.70 at 25% coverage. The signal is therefore plausibly useful for abstention at low coverage, but we make no claim that it is calibrated or that these thresholds transfer.

Retrieval-time diversity, by contrast, does *not* predict failure: its correlation with eventual correctness is weak and inconsistent in sign ( $r = 0.12$  for MC,  $r = 0.24$  for freeform), if anything with *higher* diversity slightly associating with *correct* answers — the opposite of a “high diversity means hard query” story. Whatever makes a query hard for this pipeline is determined at localization and generation, not in how differently four models phrase a hypothetical answer to the raw question.

## 5.3 Per-question-type breakdown

Table 4 slices the predicted-evidence rung by task family and evidence type. Per-cell  $n$  is single-digit, so these are point estimates, indicative only.

Two patterns stand out. Single-paper questions localize far better than multi-paper (evidence-F1 0.372 vs. 0.078): the coarse key credits a multi-paper match only if *every* constituent page is right, so per-paper error compounds. And citation-context evidence scores 0.000, matching the reference-index failure category — a semantic page-ranker has no enumerated-reference signal. Both are limitations of the current localizer, not the diagnostic protocol.

## 6 Conclusion

Read as a diagnostic rather than a leaderboard entry, the study supports a few claims with some confidence. Knowing the right paper is far from knowing where in it the evidence lies: evidence-

F1 given gold papers is only 0.217, and MC accuracy given the gold value plus page context (0.829) collapses to 0.415 once localization is left to the system — a gap surviving paired testing and a matched-serialization control. Multi-paper retrieval (Recall@5 0.150) is a larger drag than our original framing suggested, and the single-locator architecture caps multi-paper evidence-F1 by construction; the two infrastructure findings (canonical-vs-arXiv drift, OpenReview fetch failure) are real and venue-correlated. Other results remain exploratory: CMHA’s retrieval gains are not significant against the strongest baselines, the semantic-entropy ensemble does not beat the best single generator on the gold-value rung, and generation-time entropy predicts errors only weakly (AUROC 0.65/0.74, chance-spanning CIs) — though abstaining unless all four models agree is a usable low-coverage operating point. A held-out evaluation and matched serialization for the gold-evidence rung would most sharpen these conclusions.

## 7 Related Work

**Retrieval and uncertainty.** RAG (Lewis et al., 2020) grounds generation in retrieved passages; HyDE (Gao et al., 2023a) anchors dense retrieval on a hypothetical answer — we ensemble HyDE across models, on paper metadata rather than passages. Semantic entropy (Kuhn et al., 2023) clusters generations by bidirectional entailment for a hallucination signal; we apply the same criterion across models, as both an aggregation rule and a confidence score. Deciding when to abstain is the subject of selective prediction (Kamath et al., 2020; Geifman and El-Yaniv, 2017); our abstain-unless-unanimous point is a coarse, uncalibrated instance.

**Literature-grounded and scientific QA.** QASPER (Dasigi et al., 2021) pairs questions with sentence-level evidence in one paper; SciFact (Wadden et al., 2020) verifies claims against abstracts; PaperQA (Lála et al., 2023), closest to us, is a retrieval-then-answer agent over the literature. LITTRACEQA adds a checkable evidence span at page/label granularity and a 27,487-paper pool; answering from its tables is the table-QA and document-parsing problem (Zhong et al., 2020; Herzig et al., 2020; Chen et al., 2020; Zhu et al., 2021), its multi-paper questions the multi-document-synthesis problem

(Caciularu et al., 2023).

**Attribution and oracle ablation.** ALCE (Gao et al., 2023b) and FActScore (Min et al., 2023) score whether *generated text* is attributable; our metric instead scores the system’s own *locator* against a gold page/label, extending the retriever-vs-reader oracle logic of DrQA (Chen et al., 2017).

## 8 Limitations

The public development set has 55 questions, and it is also the set we inspected while designing the pipeline. Prompt wording, the localizer’s caption-matching rules and evidence-type gate, the NLI threshold, and context lengths were all chosen by looking at these questions; there is no held-out split, so the bootstrap CIs we report capture sampling noise but not this adaptation, and every number here should be read as diagnostic rather than as an estimate of blind performance. Stratifying by task family (26 single-paper / 29 multi-paper) or evidence type (17/15/9/7/7 across table/figure/text/citation/equation) yields single-digit per-cell counts, so per-stratum comparisons are indicative at best.

The gold-value generation rung feeds the model the benchmark’s short `evidence_text_or_value` string, not a reconstructed table, a full page, or a figure image, and it uses a different serialization and token budget from the predicted-evidence rung. It therefore bounds the localization→generation loss rather than isolating intrinsic generation error; a rung with manually verified, answer-sufficient evidence (and matched serialization) would give a cleaner estimate and is left to future work. All multiple-choice numbers additionally assume access to gold option text, which is not in the public input-only schema; a genuine blind submission would need to generate candidate options itself. Our table-answer aggregation is a simple longest-non-empty heuristic, not a principled multi-model table-merging method. Our localization method only recognizes bare numbered Table/Figure captions and does not handle multi-page tables, sub-panel figure references, or evidence that spans multiple locations. We initially bounded the end-to-end rung to CMHA’s top-1 paper (28 additional PDFs) before completing the top-5 setting (238 total unique candidates) once fetched (§5); their paper-F1 numbers are not directly comparable (different-sized predicted sets), though evidence-F1/MC/EM are.

Similarly, our systematic arXiv-vs-canonical ablation (§2.1) covers only the 52 of 70 gold papers we could confidently match to an arXiv entry by title, and could not include our original hand-verified example (ac12025\_00005) due to a title-matching failure on that specific record. Finally, our OpenReview-fetch workaround (§2.2) is a manual, rate-limited, semi-automated procedure rather than a general, robust solution: it demonstrates the failure mode and one way through it, not a production-ready fix.

## 9 Ethics Statement

This study involves no human subjects and no personal data. All source papers are publicly available academic publications, accessed via their official venue-hosted or organizer-provided URLs, respecting observed rate limits. We report the OpenReview bot-challenge workaround (§2.2) purely as a reproducibility/fairness concern flagging that venue-correlated access barriers can confound evaluation, not as a method for circumventing access controls at scale.

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## A Reproducibility Details

**Models.** All LLM calls go through ASU’s OpenAI-compatible “Voyager” gateway. Retrieval hypotheses and generation answers both use the same four models: qwen3-235b-a22b-instruct-2507, llama4-maverick-17b, gemma4-31b-it, and granite41-30b. Hypothesis generation uses temperature 0.7, max\_tokens= 256; answer generation uses temperature 0.0 (greedy), max\_tokens= 400 (freeform/MC) or 1200 (table);

3 retries on API error. No other decoding parameters are set (provider defaults for `top_p` etc.). We do not control a sampling seed; temperature-0 answering is close to deterministic but the provider does not guarantee bitwise reproducibility.

**Embeddings and indices.** Both paper retrieval and within-paper page/caption ranking embed text with BAAI/bge-large-en-v1.5 (transformers, mean-pooled,  $L_2$ -normalized; fp16 on GPU). The paper index is a FAISS IndexFlatIP (exact inner product, i.e. cosine on normalized vectors) with an IndexIDMap, over title+abstract embeddings of all 27,487 pool papers, built at batch size 256.

**PDF parsing.** Canonical PDFs are parsed with PyMuPDF (pymupdf); pages are 1-indexed to match the gold page field (verified on ac12025\_00005). Caption detection uses a case-insensitive regex matching a Table or Figure token, whitespace, a number with an optional trailing letter (captured), and a colon or period (full pattern in `src/fetch/pdf_parse.py`). For the predicted-evidence and matched-context rungs, the located page’s extracted text is sliced to 1500 characters and then re-capped to 1200 by the prompt builder; that 1200-character window is what the model sees per evidence item.

**NLI clustering.** Cross-model freeform answers are clustered with cross-encoder/nli-deberta-v3-base (run on CPU). An ordered pair counts as entailment when the softmax entailment probability is  $\geq 0.5$ ; two answers are equivalent if their normalized strings match or entailment holds both ways; equivalence is closed transitively by union-find. The final answer is the lowest-index member of the largest cluster. Multiple-choice “clustering” is exact-letter grouping.

**Statistics.** Bootstrap CIs resample the 55 query ids with replacement, 2000 draws (rung comparison uses 1000 unless noted), percentile method, fixed seed 0. Paired comparisons use a percentile bootstrap on the per-query metric difference (continuous metrics) or McNemar’s test on per-query correctness (binary metrics). AUROC is the tie-aware Mann–Whitney statistic; its CI is a 2000-draw bootstrap.

**Prompt.** The generation prompt is fixed across models and rungs: a one-line instruction to answer only from the evidence and not fabricate specifics,

| Retrieval config            | Hit@5        | R@5          | MRR          | SetR@5       |
|-----------------------------|--------------|--------------|--------------|--------------|
| CMHA (ours)                 | <b>0.636</b> | <b>0.461</b> | <b>0.492</b> | <b>0.382</b> |
| Single HyDE                 | 0.582        | 0.407        | 0.486        | 0.327        |
| Same-model $\times 4$       | 0.545        | 0.391        | 0.452        | 0.309        |
| BM25                        | 0.564        | 0.405        | 0.436        | 0.345        |
| Dense (raw)                 | 0.527        | 0.396        | 0.450        | 0.345        |
| <i>CMHA, by task family</i> |              |              |              |              |
| single ( $n=26$ )           | 0.808        | 0.808        | 0.703        | 0.808        |
| multi ( $n=29$ )            | 0.483        | 0.150        | 0.302        | 0.000        |

Table 5: Retrieval-availability metrics, top-5.

| Strategy             | Ev.-F1       | T/F <sub>32</sub> | txt/cit/eq <sub>23</sub> |
|----------------------|--------------|-------------------|--------------------------|
| Ours (caption + sim) | <b>0.217</b> | <b>0.294</b>      | 0.109                    |
| Whole-page dense     | 0.203        | 0.271             | 0.109                    |
| BM25 over pages      | 0.118        | 0.172             | 0.043                    |
| Page-1 floor         | 0.000        | 0.000             | 0.000                    |

Table 6: Localization-strategy comparison on the oracle-papers rung (evidence-F1, macro over 55 questions), split by whether the evidence type carries a numbered label (T/F,  $n=32$ ) or not ( $n=23$ ).

the numbered evidence excerpts, the question, a JSON schema line for the required answer type(s), and (for MC) the gold option letters and text, (for table) the column names and types. It is reproduced verbatim in `src/generation/prompts.py` in our implementation.

## B Retrieval and Localization Diagnostics

**Retrieval availability.** Table 5 re-scores the Table 1 predictions with availability-oriented metrics: Hit@5 (a gold paper is in the top 5), Recall@5 (fraction of gold papers retrieved), MRR, and SetRecall@5 (the full gold set is in the top 5), overall and split by task family. Multi-paper Recall@5 (0.150) is the binding constraint.

**Localization strategies.** Table 6 compares four within-paper localization strategies on the oracle-papers rung, scored on evidence-F1 (macro over the 55 dev questions, the official coarse key), split by whether the evidence type carries a numbered label. “Ours” is the shipped filter-then-rank localizer and reproduces the ladder’s Localization rung (Table 2) exactly. It beats plain whole-page dense ranking by only +0.014 overall (+0.024 on table/figure questions, where the caption logic applies; the two are identical on unlabelled types by construction). BM25 over pages is far worse, so the embedding similarity, not the caption filter, is carrying the signal. Every strategy has a low ceiling here.

## C Full Per-Question Outcomes

Table 7 reports, for all 55 dev-set questions, whether the localization step returned the exact gold evidence locator (*Loc.*) and whether the final answer was correct (*Corr.*) in the predicted-evidence generation rung, alongside task family and primary evidence type. This is the full, mechanically-derived population underlying the stratified  $n = 20$  sample in Table 3 (§5.1); it is provided here for full transparency and is not itself hand-labeled.

| Query | Family | Evid.    | Loc. | Corr. | Query | Family | Evid.    | Loc. | Corr. | Query | Family | Evid.    | Loc. | Corr. |
|-------|--------|----------|------|-------|-------|--------|----------|------|-------|-------|--------|----------|------|-------|
| q_001 | single | table    | N    | N     | q_021 | single | table    | N    | N     | q_040 | multi  | equation | N    | N     |
| q_002 | single | text     | N    | N     | q_022 | multi  | equation | N    | N     | q_041 | multi  | equation | N    | Y     |
| q_003 | single | citation | N    | N     | q_023 | multi  | citation | N    | N     | q_042 | multi  | citation | N    | Y     |
| q_004 | single | figure   | N    | N     | q_024 | single | figure   | N    | N     | q_043 | multi  | table    | N    | Y     |
| q_005 | single | citation | N    | N     | q_025 | multi  | text     | N    | N     | q_044 | multi  | table    | N    | N     |
| q_006 | single | figure   | N    | N     | q_026 | single | figure   | Y    | Y     | q_045 | multi  | table    | N    | Y     |
| q_007 | single | figure   | Y    | Y     | q_027 | single | citation | N    | N     | q_046 | multi  | figure   | N    | N     |
| q_008 | single | text     | Y    | Y     | q_028 | multi  | table    | N    | N     | q_047 | multi  | figure   | N    | N     |
| q_009 | single | table    | Y    | N     | q_029 | multi  | table    | N    | N     | q_048 | multi  | figure   | N    | Y     |
| q_010 | single | table    | Y    | N     | q_030 | multi  | table    | N    | N     | q_049 | multi  | text     | N    | N     |
| q_011 | single | table    | Y    | Y     | q_031 | multi  | figure   | N    | N     | q_050 | multi  | equation | N    | Y     |
| q_012 | single | table    | N    | N     | q_032 | multi  | figure   | N    | Y     | q_051 | multi  | equation | N    | Y     |
| q_013 | single | figure   | N    | N     | q_033 | multi  | figure   | N    | Y     | q_052 | single | table    | N    | N     |
| q_014 | single | table    | N    | N     | q_034 | multi  | figure   | N    | N     | q_053 | single | table    | Y    | N     |
| q_015 | single | text     | N    | Y     | q_035 | multi  | text     | N    | Y     | q_054 | single | table    | N    | N     |
| q_017 | single | citation | N    | N     | q_036 | multi  | text     | N    | N     | q_055 | single | figure   | Y    | N     |
| q_018 | single | equation | Y    | Y     | q_037 | multi  | text     | N    | Y     | q_056 | multi  | table    | N    | N     |
| q_019 | single | citation | N    | N     | q_038 | multi  | text     | N    | N     |       |        |          |      |       |
| q_020 | multi  | figure   | N    | N     | q_039 | multi  | equation | N    | N     |       |        |          |      |       |

Table 7: Per-question localization and final-answer correctness in the predicted-evidence generation rung, mechanically derived from `results/diagnostic/failure_dossier.jsonl`, for all 55 dev-set questions (three column-groups of the same five fields, read left to right, top to bottom). This is the full population underlying the stratified  $n = 20$  sample in Table 3 (§5.1); it is mechanically derived, not itself hand-labeled. Query IDs skip q\_016: it is simply absent from the organizer-released public validation `.jsonl` (55 records total, non-contiguous IDs); we did not exclude it ourselves and do not know the organizers’ reason.